How-To Library

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Major and Minor Relationships in Test Data Management

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Abstract

This document explains how major relationships and minor relationships work between tables in an entity in Test Data Management (TDM). It assumes that you have knowledge of the data subset feature in TDM. With sample data, the document explains how the relationship assigned to tables and the filter assignment can impact the output in a data subset operation.

Supported Versions

- Test Data Management 9.5.1 to 9.6.0

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Overview

You create entities in TDM to perform data subset operations across related tables. Entities ensure that all related tables are included in the subset operation and the subset data remains referentially intact.

A driving table is the first table that you include in an entity. When you add a driving table, TDM adds all tables related to the driving table to the entity. These tables are the related tables. The driving table and related tables might have parent-child relationships or constraints based on the existence of primary and foreign keys. So a driving table might be the child table of another table in the entity. The driving table might also be the parent table of other tables in the entity. After you select the tables, you define filters for columns in the tables to provide the subset criteria.

Tables in an entity might be related by physical or logical constraints. Constraints define the parent-child relationship. The constraint might be a primary key constraint that is from the data source. The constraint might be a logical constraint. A logical constraint defines a parent-child relationship based on columns that are not keys in the data source.

Within the entity, all constraints are assigned a relationship severity of either major or minor. The relationship between the tables determines the tables from which the subset filter selects data and consequently determines the output data. You can change the relationship assigned to table constraints.

Major and Minor Relationships

Major and minor relationships determine the output of a subset operation. You can change the severity to change the final output of a subset operation.

Major relationships are relationships that move in both directions – child to parent and parent to child or bottom to top and top to bottom. When you want the parent table to select additional child records that match the parent, assign a major relationship between two tables.

Minor relationships are unidirectional relationships, from child to parent, or from bottom to top. Minor relationships do not move in the reverse direction from parent to child. When you do not want the parent table to select additional child table records that match the parent, assign a minor relationship between two tables.

Relationship Severity and Subset Output

When you create an entity, TDM ensures that you include all related tables in a subset operation, thereby avoiding any referential integrity issues in the subset data. The tables within the entity are assigned relationship severities. The relationship severity determines whether the subset operation can move from child to parent or both from child to parent and parent to child. Minor relationships move from child to parent, and major relationships move in both directions. The relationship severity can therefore impact the output of the subset operation.

Because the direction of movement of the operation is restricted by the relationship that you assign between tables, you must also consider the point from which the subset operation begins. A subset operation begins from the table on which you apply the filter. Based on the relationship between the tables, the operation might not include all tables in the entity. The subset output therefore depends on the relationships between the tables and the table you apply the filter on.

The following scenarios show the types of entity relationships:

- All table relationships are major.
- The relationship between Table 1 and Table 2 is minor. The other relationships are major.
- All table relationships are minor.

Each scenario shows the impact of changing the filter table and the impact of relationships and filters on data subset output. The combination of relationships between tables and the subset filter applied decide the data that the subset operation picks up.

Sample Data

In this sample data set, Table 1 is the parent table of Table 2 and Table 3. Table 3 is the parent table of Table 2.

With this sample data, you perform a subset operation to filter records with the year 2013.

Table 1

Year
2012
2013

Table 2

ID_NUM	Year
1	2012
2	2012
3	2012
4	2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

Scenario: All Table Relationships are Major

Assign a major relationship between all the three tables. Keep the relationships constant, and perform the subset operation applying the filter on a different table each time.

Table 1 has the Filter

Because Table 1 has the filter, the subset operation starts in Table 1.

The subset filter selects one matching record from Table 1. The relationship between Table 1 and Table 2 is major and moves in both directions. The subset filter selects the one matching record in Table 2, moving from parent to child. The subset filter picks up the two matching records from Table 3. There is a major relationship between tables 2 and 3, so the filter selects another matching record from Table 2, moving from parent to child Table 3 to Table 2. The filter operation selects another record from Table 1, moving from child to parent, Table 2 to Table 1. There is a major relationship between tables 1 and 2, so the filter selects the other two rows from Table 2 moving from parent to child.

As a result, the filter selects more than the filter criteria data in the subset. The operation moves in both directions and the output includes all related rows. The following tables describe the output of the subset operation:

Table 1

Year
2012
2013

Table 2

ID_NUM	Year
1	2012

ID_NUM	Year
2	2012
3	2012
4	2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

Table 2 or Table 3 has the Filter

In this example, both these cases have the same result as when you apply the filter to Table 1. All of the relationships are major, and both child to parent and parent to child movement is possible. Regardless of the table that the subset operation begins in, the operation moves in both directions and the output includes all related rows.

Scenario: Relationship between Table 1 and 2 is Minor

Assign a minor relationship between Table 1 and Table 2. Assign a major relationship between Table 2 and Table 3 and between Table 3 and Table 1. Keep the relationships constant, and perform the subset operation applying the filter on a different table each time.

Table 1 has the Filter

Because Table 1 has the filter, the subset operation starts in Table 1.

The subset filter selects one matching record from Table 1. There is a major relationship between Table 1 and Table 3. The filter selects two rows from Table 3 moving from parent to child. There is a minor relationship between Table 1 and Table 2, which does not allow parent to child movement. Tables 2 and 3 have a major relationship. The filter selects two related records from Table 2, moving from parent to child, Table 3 to Table 2. The filter selects another matching record from Table 1 because Table 1 and Table 2 have a minor relationship which is movement from child to parent tables.

The output in this scenario includes an additional row, based on the filter criteria. The following tables describe the output of the subset operation:

Table 1

Year
2012
2013

Table 2

ID_NUM	Year
3	2012
4	2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

Table 2 has the Filter

Because Table 2 has the filter, the subset operation starts in Table 2.

The subset filter selects one row from Table 2. There is a minor relationship between Table 1 and Table 2. The filter selects one row from Table 1 moving from child to parent. Table 1 and Table 3 have a major relationship. The filter selects two rows from Table 3 moving from parent to child. Because the tables have a major relationship, the filter selects one row from Table 2 moving from Table 3 to Table 2. The filter selects one row from Table 1 moving from Table 2 to Table 1 as the tables have a minor relationship and can move from child to parent tables.

The following tables describe the output of the subset operation:

Table 1

Year
2012
2013

Table 2

ID_NUM	Year
3	2012
4	2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

In this example, the final output of the subset operation is the same as before, even though the subset operation moves in a different sequence between tables.

Table 3 has the Filter

Because Table 3 has the filter, the subset operation starts in Table 3.

The filter selects two rows from Table 3. There is a major relationship between Table 3 and Table 2, so the filter selects two rows from Table 2 moving from parent to child. The filter selects two rows from Table 1 moving from Table 2 to Table 1 because there is a minor relationship between the tables which is movement from child to parent tables.

In this example, the output of the subset operation is the same as before, even though the subset operation moves in a different sequence between tables. The following tables describe the output of the subset operation:

Table 1

Year
2012
2013

Table 2

ID_NUM	Year
3	2012
4	2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

In this example, with the one relationship minor and the rest major, changing the table on which you apply the filter does not vary the output.

Scenario: All Table Relationships are Minor

Assign a minor relationship between all the three tables. Keep the relationships constant, and perform the subset operation applying the filter on a different table each time.

Table 1 has the Filter

Because Table 1 has the filter, the subset operation starts from Table 1.

The filter selects one matching row from Table 1. Minor relationships move only from child to parent, so the filter does not consider or select any rows from Table 2 or Table 3.The following table describes the output of the subset operation:

Table 1

Year
2013

Table 2 has the Filter

Because Table 2 has the filter, the subset operation starts from Table 2.

The filter selects one row from Table 2 and one row from Table 1 moving from child to parent, Table 2 to Table 1. The filter selects one row from Table 3 moving from child to parent Table 2 to Table 3.

The following tables describe the output of the subset operation:

Table 1

Year
2013

Table 2

ID_NUM	Year
4	2013

Table 3

ID_Parent	Year	ID_Parent
2	2013	4

Table 3 has the Filter

Because Table 3 has the filter, the subset operation starts from Table 3.

The filter selects two rows from Table 3 and one row from Table 1 moving from child to parent, Table 3 to Table 1. The filter does not select any rows from Table 2 because Table 2 and Table 3 have a minor relationship that does not include movement from parent to child tables.

The following tables describe the output of the subset operation:

Table 1

Year
2013

Table 3

ID_NUM	Year	ID_Parent
1	2013	3
2	2013	4

In this example, without changing the relationship between tables, the subset output varies each time you apply the filter on a different table. With a different set of relationship severities, changing the table on which you apply the filter might not impact the subset output.

Conclusion

The different scenarios return different results. The operation moves in a different order based on the relationship between the tables and the table on which you apply the filter. Changing the relationship between tables can impact the data subset output. Changing the table on which you apply the filter can also impact the output.

For the three tables in the sample data, there are eight possible permutations of relationships between the tables. If you apply the filter on a different table each time, there are 24 different ways to configure parameters for the same subset operation. Each time the output might differ, based on which table you apply the filter on and the relationship of the table with the other tables.

If the subset operation includes more tables, the number of possible options increases further. It is important to check which relationship or combination of relationships between tables and filter criteria can provide the best output to meet your subset requirements.

Authors

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